Assignment 4

Sharded Key/Value Service

Arjun Ahuja
M.S. in Computer Science

Overview

- Assignment 3: Distributed Key/Value Database using Paxos.
- Assignment 4: “sharded” key/value store.
- Assignment is divided into two parts:
    - Design a Shard Master
    - Design a sharded Key Value Database.

Sharding Recap

- A shard is a subset of the key/value pairs stored in the overall key-value store.
- Carefully split the data (and the requests accessing it) across many servers.
- Advantages? Scalability (Horizontal)
- For example, if our key space is the integers[0, 100], the keys in the range[0, 50] might be one shard, the keys [51, 60] another shard, etc.
- Use case: Sites like Facebook, Airbnb, that have so many users that no single server can store all their data, or handle all the requests for data.

Replica Groups

- Each shard is managed by a handful of servers (Replication). These handful of servers comprise a replica group.
- A replica group can manage multiple shards.
- Use Paxos to decide on the operations to execute within this replica group.
- Operations to execute can be Put/Get/PutHash/ReConfiguration.

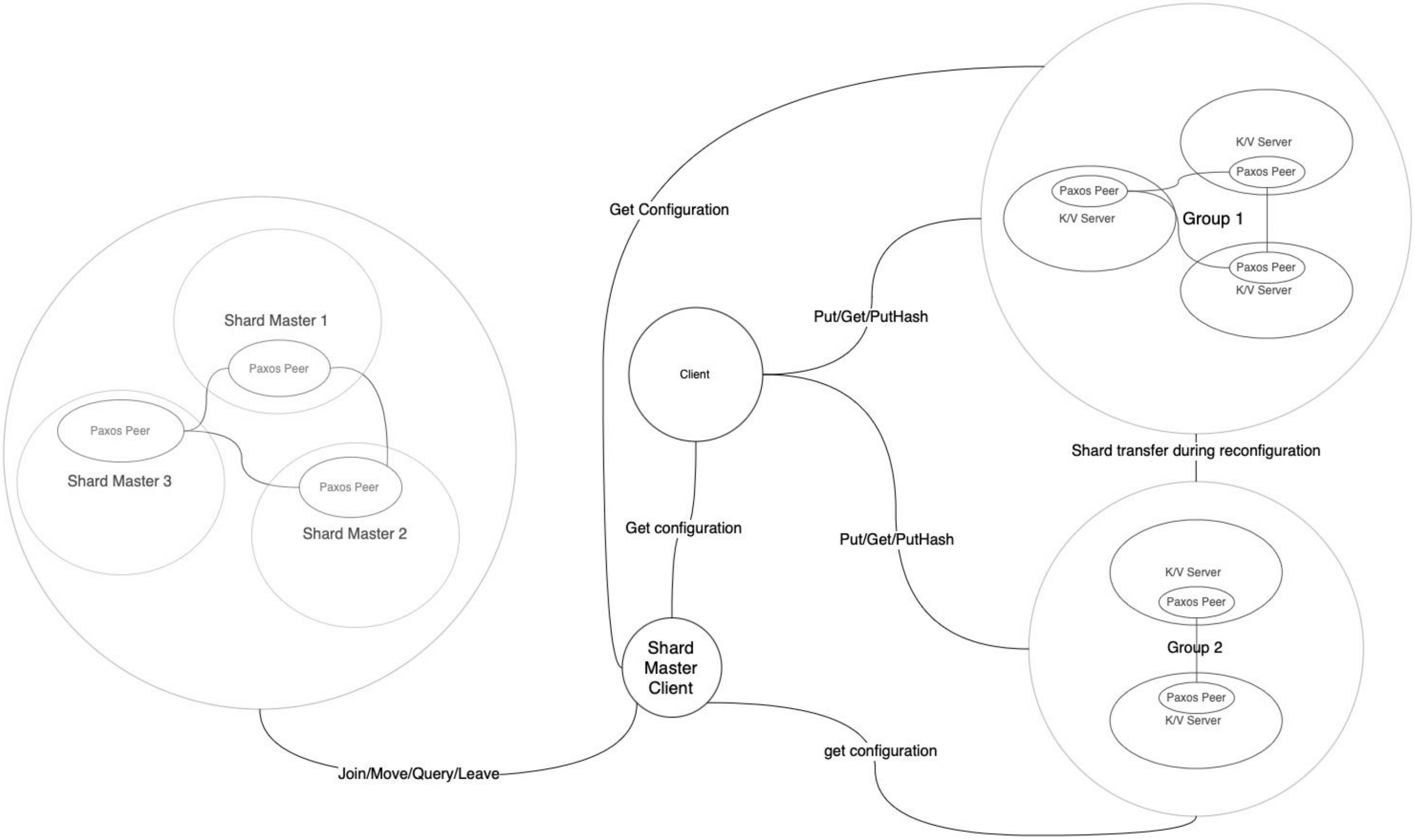
Shard Master

- The shardmaster manages a sequence of numbered configurations.
- Each configuration describes a set of replica groups and an assignment of shards to replica groups.
- Clients and Servers contact the shardmaster when they want to know the current (or a past) configuration.
- There can be multiple servers in a shard master.
- Load Balance the configurations (not on the data but rather on shards.)
- To implement: Join, Leave, Move, and Query.

Load Balance Example

- Initially Configuration: shard 1, shard 2 -> group 1, shard 3, shard 4 -> group 2.
- New Group group 3 joins.
- Shards move such that final configuration is load balanced:
  - Final Configuration: shard 1 -> group 1, shard 3, shard 4 -> group 2, shard 2 -> group 3.
  - More possible possible configuration:
    - shard 2 -> group 1, shard 3, shard 4 -> group 2, shard 2 -> group 3.
    - shard 1, shard 2 -> group 1, shard 3 -> group 2, shard 4 -> group 3.
  - Not optimal configuration:
    - shard 3 -> group 1, shard 4 -> group 2, shard 1, shard 2 -> group 3.

Design



Get Configuration
Shard Master 1
Paxos Peer
Paxos Peer
Shard Master 3
Paxos Peer
Shard Master 2
Client
Get configuration
Shard Master Client
Join/Move/Query/Leave
Put/Get/PutHash
Put/Get/PutHash
get configuration
K/V Server
Paxos Peer
Paxos Peer
K/V Server
Group 1
Paxos Peer
K/V Server
Shard transfer during reconfiguration
K/V Server
Paxos Peer
Group 2
Paxos Peer
K/V Server

Questions